SVM Prediction on Telco Customer Churn Dataset

The Support Vector Machine (SVM) algorithm was applied to the Telco Customer Churn dataset to predict whether a customer is likely to leave the telecom service or continue using it. The model was trained using customer demographic information, account details, service subscriptions, and billing-related features.

After training, the SVM model analyzed customer behavior patterns and classified customers into two categories:

- **Churn (Yes):** Customers who are likely to discontinue the service.
- **Churn (No):** Customers who are likely to remain with the company.

The prediction results indicate that customers with shorter tenure, higher monthly charges, and certain service-related issues are more likely to churn. On the other hand, customers with longer service periods, stable contracts, and multiple subscribed services tend to remain loyal to the company.

The SVM model successfully identified potential churn customers with good accuracy, making it a valuable tool for customer retention strategies. By predicting churn in advance, telecom companies can take proactive measures such as offering personalized discounts, improved customer support, loyalty programs, or special service plans to retain at-risk customers.

Predicted Outcome

Based on the analysis using SVM:

- The majority of customers were predicted as **non-churn customers**, indicating they are likely to continue using the service.
- A smaller group of customers was predicted as **potential churn customers**, requiring attention from the company.
- The model demonstrated that customer tenure, contract type, monthly charges, and total charges are among the most influential factors affecting churn prediction.

Conclusion

The SVM-based prediction model effectively classifies customers according to their likelihood of churning. The prediction results can help the organization identify high-risk customers early and implement targeted retention strategies, ultimately reducing customer loss and improving business profitability. This predictive approach supports data-driven decision-making and enhances customer relationship management within the telecom industry.